

Shivankur Ghaie

Capstone Project - 1

1. Creation of storage account:

Create a storage account ...

manage your storage account together with other resources.

Subscription *

Resource group * [Create new](#)

Instance details

Storage account name *

Region * [Deploy to an Azure Extended Zone](#)

Primary service

Performance * ☒ Standard: Recommended for most scenarios (general-purpose v2 account)
☐ Premium: Recommended for scenarios that require low latency.

Redundancy *

[Home](#) >

 **shivghaie_1756097810267** | Overview  ...

   Delete  Cancel  Redeploy  Download  Refresh

Overview

-  Inputs
-  Outputs
-  Template

✓ Your deployment is complete

 Deployment name: shivghaie_17560978... Start time: 8/25/2025, 10:27:07 AM
Subscription: [LabsKraft](#) Correlation ID: 2a06e6b2-0d76-449d-b29f-2879c6baf10c
Resource group: [LabsKraft2027](#)

Deployment details

Next steps

[Go to resource](#)

Give feedback

 Tell us about your experience with deployment

✓ **Deployment succeeded** 

Deployment 'shivghaie_1756097810267' to resource group 'LabsKraft2027' was successful.

[Go to resource](#)

[Pin to dashboard](#)



Cost Management

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[Set up cost alerts >](#)



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shivghaie

Storage account

Enhance the security of this storage account

Create an LCM rule for my storage account

Analyze connectivity to this storage account

Search

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Partner solutions

Resource visualizer

Data storage

Upload

Open in Explorer

Delete

Move

Refresh

Open in mobile

CLI / PS

Feedback

Essentials

Resource group (move)

LabsKraft2027

Location

southindia

Subscription (move)

LabsKraft

Subscription ID

d008c7cc-9795-4292-bf79-74ae52cda63d

Disk state

Available

Tags (edit)

Add tags

Performance

Standard

Replication

Locally-redundant storage (LRS)

Account kind

StorageV2 (general purpose v2)

Provisioning state

Succeeded

Created

8/25/2025, 10:27:11 AM

JSON View

shivghaie | Containers

Storage account

Search

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Partner solutions

Resource visualizer

Add container

Upload

Refresh

Delete

Change access level

Restore containers

Edit columns

Search containers by prefix

Only show active containers

Showing all 2 items

☐	Name	Last modified	Anonymous access level	Lease state	
☐	\$logs	8/25/2025, 10:27:52 AM	Private	Available	...
☒	raw	8/25/2025, 10:28:24 AM	Private	Available	...

raw

Container

Search

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Add Directory

Upload

Refresh

Delete

Copy

Paste

Rename

Acquire lease

Break lease

Edit columns

raw

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Only show active objects

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state	
☐	retail_iot_data (...)	8/25/2025, 10:31:08 AM	Hot (Inferred)	Block blob	130.76 KiB	Available	...

2. Creation of access connector:

Create an Access Connector for Azure Databricks

Basics Tags Managed Identity Review + create

The Azure Databricks Access Connector lets you connect managed identities to an Azure Databricks account for the purpose of accessing data registered in Unity Catalog.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ

LabsKraft

Resource group * ⓘ

LabsKraft2027

Create new

Instance details

Name * ⓘ

shiv-connector

Region * ⓘ

East US



AccessConnectorCreate-20250825103701 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : AccessConnectorCreate-20250...

Subscription : LabsKraft

Resource group : LabsKraft2027

Start time : 8/25/2025, 10:37:41 AM

Correlation ID : ee40e2aa-f3e3-4fbb-adee-c05...

> Deployment details

> Next steps

Go to resource

3. Giving access connector permission for storage:

Add role assignment

Role Members Conditions Review + assign

A role definition is a collection of permissions. You can use the built-in roles or you can create your own custom roles. [Learn more](#)

Job function roles Privileged administrator roles

Grant privileged administrator access, such as the ability to assign roles to other users.

Can a job function role with less access be used instead?

Search by role name, description, permission, or ID

Type : All

Category : All

Name	Description	Type	Category	Details
Contributor	Grants full access to manage all resources, but does not allow you to assign roles in Azure R...	BuiltInRole	General	View
Azure File Sync Administrator	Provides full access to manage all Azure File Sync (Storage Sync Service) resources.	BuiltInRole	None	View
Azure Resilience Management Drills...	Administrator Role of Azure Resilience Management Drills Service	BuiltInRole	None	View

Showing 1 - 3 of 3 results.

Add role assignment

Role Members Conditions Review + assign

Showing a filtered list of roles because your permissions include a condition. [Learn more](#)
[View my access](#)

Selected role Contributor

Assign access to
☐ User, group, or service principal
☒ Managed identity

Members + Select members

Name	Object ID
No members selected	

Description Optional

Some results might be hidden due to your ABAC condition.

Subscription * LabsKraft

Managed identity Access Connector for Azure Databricks (19)

shiv

shivam-connector /subscriptions/d008c7cc-9795-4292-bf79-74ae52cda63d/resourceGroups/LabsKra...

Selected members:
shiv-connector /subscriptions/d008c7cc-9795-4292-bf79-74ae52cda63d/resourceGroups/... [Remove](#)

Add role assignment

Role Members Conditions Review + assign

A role definition is a collection of permissions. You can use the built-in roles or you can create your own custom roles. [Learn more](#)

Job function roles Privileged administrator roles

Grant access to Azure resources based on job function, such as the ability to create virtual machines.

storage blob con

Type : All

Category : All

Name	Description	Type	Category	Details
Storage Blob Data Contributor	Allows for read, write and delete access to Azure Storage blob containers and data	BuiltInRole	Storage	View
Storage Blob Data Owner	Allows for full access to Azure Storage blob containers and data, including assigning POSIX a...	BuiltInRole	Storage	View
Storage Blob Data Reader	Allows for read access to Azure Storage blob containers and data	BuiltInRole	Storage	View

Showing 1 - 3 of 3 results.

4. Creation of Unity Catalog:

Create a new catalog

A catalog is the first layer of Unity Catalog's three-level namespace and is used to organize your data assets. [Learn more](#)

Catalog name*

Type*

Standard

Storage location

Select external location

sub/path

[Create a new external location](#)

Location in cloud storage where data for managed tables will be stored. If not specified, the location will default to the metastore root location.

CancelCreate

cap1_shiv_ext_loc

Test connection

OverviewPermissionsBrowseWorkspaces

Description

Add description

About this external location

OwnerTredence LabsKraft19

Fallback mode:

Credential	cap1_shiv_ext_loc_azuremanagedidentity_1756099064822
URL	abfss://raw-1@shivghaie.dfs.core.windows.net/
Limit to read-only use	Disabled

Catalog

Starter Warehouse Pro S

Type to search...

My organization

tred_d5

system

cap1_shiv_catalog

default

information_schema

shiv

shiv_catalog

shivam_catalog_new

shivam-catalog

shivam-catalog-project

st_capstone_retail

st-catalog

Delta Shares Received

Legacy

samples

hive_metastore

Catalog Explorer

cap1_shiv_catalog

Share Create schema

Overview Details Permissions Workspaces

Description

AI generate Add

Filter schemas 2 schemas

Name	Owner	Created at
default	tredence19@michaelkloudkraft...	Aug 25, 2025, 10:55 AM
information_schema	System user	Aug 25, 2025, 10:55 AM

About this catalog

Owner Tredence LabsKraft19

Tags

Add tags

5. Creation of Namespace and Eventhub:



Create Namespace ...

Event Hubs

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

LabsKraft

Resource group *

LabsKraft2027

Create new

Instance Details

Enter required settings for this namespace, including a price tier and configuring the number of units (capacity).

Namespace name *

shiv-event

.servicebus.windows.net

Region *

South India

Pricing tier *

Basic

Browse the available plans and their features

Throughput Units *

1



shiv-eventHub | Overview

Deployment

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template



Your deployment is complete



Deployment name : shiv-eventHub
Subscription : LabsKraft
Resource group : LabsKraft2027

Start time : 8/25/2025, 11:09:47 AM
Correlation ID : d97ddce0-b067-4cbd-8e40-e3...

> Deployment details

✓ Next steps

Go to resource

shivghaie

Event Hubs Namespace

Summarize properties of this Event Hubs Namespace.

Show me metrics for this Event Hubs Namespace.

List current consumers of this Event Hubs Namespace.

+ Event Hub Delete Refresh Give feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Data Explorer

Resource visualizer

Events

Settings

Entities

Monitoring

Automation

Help

Essentials

Resource group (move) : LabsKraft2027

Status : Succeeded

Location : Central US

Subscription (move) : LabsKraft

Subscription ID : d008c7cc-9795-4292-bf79-74ae52cda63d

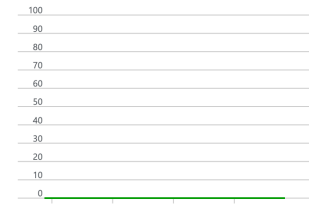
Host name : shivghaie.servicebus.windows.net

Tags (edit) : Add tags

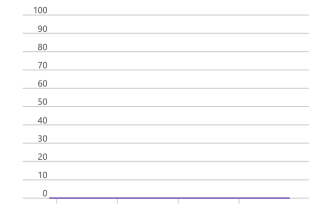
NAMESPACE CONTENTS 0 EVENT HUBS KAFKA SURFACE NOT SUPPORTED ZONE REDUNDANCY ENABLED

Show data for the last: 1 hour 6 hours 12 hours 1 day 7 days 30 days

Requests



Messages



Throughput



shiveventhub (shivghaie/shiveventhub) | Shared access policies

Event Hubs Instance

+ Add

Overview

Access control (IAM)

Diagnose and solve problems

Data Explorer

Resource visualizer

Settings

Shared access policies

Configuration

Properties

Locks

Entities

Features

Automation

Help

Search to filter items...

Policy

shive

Event Hub

Regenerate primary key Regenerate secondary key Delete

Event Hubs access control with Shared Access Signatures [Learn more](#)

Claims

☒ Manage

☐ Send

☐ Listen

Primary key

.....

Secondary key

.....

Primary connection string

.....

Secondary connection string

.....

SAS policy ARM ID

/subscriptions/d008c7cc-9795-4292-bf79-74ae52cda63d/resourceGroups/LabsKraft...

☐ Show AMQP connection strings

6. Creating Synapse Workspace:

Create Synapse workspace ...

Resource group * ⓘ

LabsKraft2027

▼

Create new

Managed resource group ⓘ

Enter managed resource group name

Workspace details

Name your workspace, select a location, and choose a primary Data Lake Storage Gen2 file system to serve as the default location for logs and job output.

Workspace name *

shiv-synapse

✓

Region *

(Asia Pacific) South India

▼

Select Data Lake Storage Gen2 * ⓘ

☒ From subscription ☐ Manually via URL

Account name * ⓘ

shivghaie

▼

Create new

File system name *

raw-1

▼

Create new

☒ Assign myself the Storage Blob Data Contributor role on the Data Lake Storage Gen2 account to interactively query it in the workspace.

Create Synapse workspace ...

* Basics * Security Networking Tags Review + create

Configure security options for your workspace.

Authentication

Choose the authentication method for access to workspace resources such as SQL pools. The authentication method can be changed later on. [Learn more](#)

Authentication method ⓘ

☒ Use both local and Microsoft Entra ID authentication

☐ Use only Microsoft Entra ID authentication

SQL Server admin login * ⓘ

shiv

✓

SQL Password ⓘ

.....

✓

Confirm password

.....

✓

System assigned managed identity permission

Select to grant the workspace network access to the Data Lake Storage Gen2 account using the workspace system identity. [Learn more](#)

Home >

shiv-synapse

Synapse workspace

Search

◊ ◀

+ New dedicated SQL pool

+ New Apache Spark pool

🔄 Refresh

🔑 Reset SQL admin password

🗑️ Delete

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Analytics pools

Security

Monitoring

Automation

Help

Essentials

Resource group (move) : [LabsKraft2027](#)

Status : Succeeded

Location : North Central US

Subscription (move) : [LabsKraft](#)

Subscription ID : d008c7cc-9795-4292-bf79-74ae52cda63d

Managed virtual network : No

Managed Identity object ... : a7ddc71e-f6af-4752-b063-e5ebd717f847

Workspace web URL : <https://web.azuresynapse.net?workspace=%2fsubscriptions%2f...>

Tags (edit) : [Add tags](#)

Networking : [Show firewall settings](#)

Primary ADLS Gen2 acco... : <https://shivghaie.dfs.core.windows.net>

Primary ADLS Gen2 file s... : raw-1

SQL admin username : shiv

SQL Microsoft Entra admin : trudence19@michaelkloudkrafthotmail.onmicrosoft.com

Dedicated SQL endpoint : shiv-synapse.sql.azuresynapse.net

Serverless SQL endpoint : shiv-synapse-ondemand.sql.azuresynapse.net

Development endpoint : <https://shiv-synapse.dev.azuresynapse.net>

JSON View

Getting started

Open Synapse Studio

Start building your fully-integrated analytics solution and unlock new insights.

Open

Read documentation

Learn how to be productive quickly. Explore concepts, tutorials, and samples.

Learn more

Analytics pools

🔍 Search to filter items...

Name	Type	Size
------	------	------

SQL pools

🔗

Build in

🔗

Scenarios

🔗

Auto

Add or remove favorites by pressing **Cmd+Shift+F**

7. Creation of new Dedicated SQL Pool:

Home > shiv-synapse >

New dedicated SQL pool

...

* Basics

* Additional settings

Tags

Review + create

Product details

Azure Synapse Analytics dedicated SQL pool by Microsoft

[Terms of use](#) | [Privacy policy](#)

Est. Cost Per Hour
1256.24 INR

[View pricing details](#)

Terms

By clicking Create, I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. For additional details see [Azure Marketplace Terms](#).

Basics

Dedicated SQL pool name : shiv_sql_pool

Performance level : DW1000c

Additional settings

Use existing data : None

Collation : SQL_Latin1_General_CP1_CI_AS

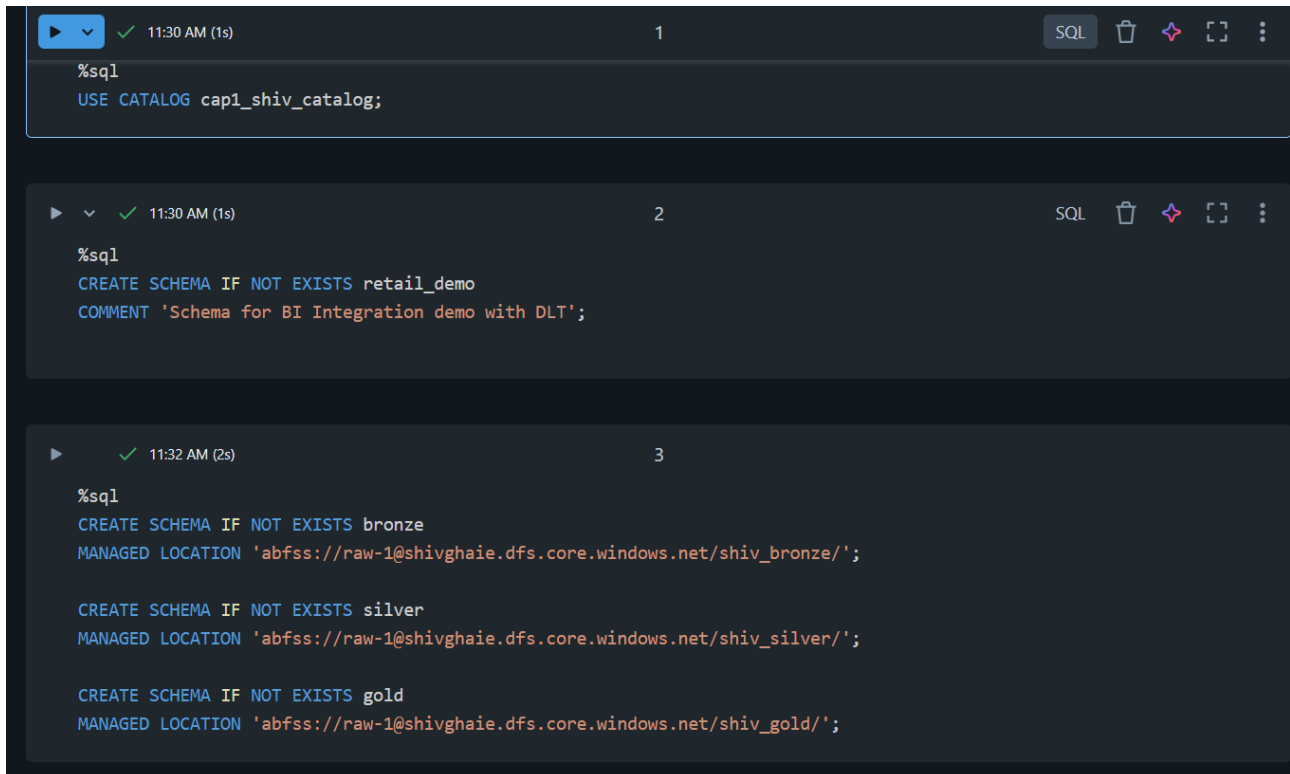
Tags

Ingestion Mode

1. ADLS

Made a new Notebook in my workspace and wrote code for Medallion Architecture

(i) Schema



The screenshot shows a Databricks notebook interface with three SQL cells. The first cell sets the catalog to 'cap1_shiv_catalog'. The second cell creates a schema named 'retail_demo' with a comment. The third cell creates three schemas: 'bronze', 'silver', and 'gold', each with a managed location in the ADLS.

```
%sql
USE CATALOG cap1_shiv_catalog;

-- Cell 2
%sql
CREATE SCHEMA IF NOT EXISTS retail_demo
COMMENT 'Schema for BI Integration demo with DLT';

-- Cell 3
%sql
CREATE SCHEMA IF NOT EXISTS bronze
MANAGED LOCATION 'abfss://raw-1@shivghaie.dfs.core.windows.net/shiv_bronze/';

CREATE SCHEMA IF NOT EXISTS silver
MANAGED LOCATION 'abfss://raw-1@shivghaie.dfs.core.windows.net/shiv_silver/';

CREATE SCHEMA IF NOT EXISTS gold
MANAGED LOCATION 'abfss://raw-1@shivghaie.dfs.core.windows.net/shiv_gold/';
```

(ii) Bronze Layer

```
from dlt import table
import pyspark.sql.functions as F
from pyspark.sql.types import StructType, StructField, StringType, DoubleType, IntegerType,
TimestampType
# :white_check_mark: Define schema for retail IoT CSV
iot_schema = StructType([
    StructField("OrderID", StringType(), True),
    StructField("CustomerID", StringType(), True),
    StructField("Region", StringType(), True),
    StructField("StoreID", StringType(), True),
    StructField("Product", StringType(), True),
    StructField("Quantity", IntegerType(), True),
    StructField("UnitPrice", DoubleType(), True),
    StructField("SalesAmount", DoubleType(), True),
    StructField("OrderTime", StringType(), True),
```

```

    StructField("DeviceType", StringType(), True),
    StructField("Temperature", DoubleType(), True),
    StructField("DeviceStatus", StringType(), True),
    StructField("AnomalyFlag", IntegerType(), True),
    StructField("AnomalyType", StringType(), True)
]
# :white_check_mark: Paths (use subdirectories to avoid overlap with DLT system files)
raw_path = "abfss://raw-1@shivghaie.dfs.core.windows.net/raw_data/"
schema_path = "abfss://raw-1@shivghaie.dfs.core.windows.net/dlt_schemas/bronze_orders"
checkpoint_path = "abfss://raw-1@shivghaie.dfs.core.windows.net/dlt_checkpoints/bronze_orders"
@table(
    name="cap1_shiv_catalog.capstone_retail_shivankur.bronze_orders",
    comment="Bronze table - raw ingested IoT retail data via Auto Loader"
)
def bronze_orders():
    return (
        spark.readStream
            .format("cloudFiles")
            .option("cloudFiles.format", "csv")
            .option("header", True)
            .option("cloudFiles.schemaLocation", schema_path)
            .schema(iot_schema) # :white_check_mark: manually set schema (no infer needed)
            .load(raw_path)     # :white_check_mark: points to /raw_data/ folder
    )

```

(iii) Silver Layer

```

# Silver Layer - Cleaned IoT Orders
@table(
    name="capstone_retail_shivankur.silver_orders",
    comment="Silver layer: cleaned and deduplicated IoT orders with anomaly handling"
)
def silver_orders():
    # Read Bronze
    orders = dlt.read("cap1_shiv_catalog.capstone_retail_shivankur.bronze_orders")
    # Standardize schema
    orders = (
        orders.withColumn("Quantity", F.col("Quantity").cast(IntegerType()))
        .withColumn("UnitPrice", F.col("UnitPrice").cast(DoubleType()))
        .withColumn("SalesAmount", F.col("SalesAmount").cast(DoubleType()))
        .withColumn("Temperature", F.col("Temperature").cast(DoubleType()))
        .withColumn("OrderTime", F.to_timestamp("OrderTime", "M/d/yyyy H:mm"))
    )
    # Deduplicate by OrderID
    orders = orders.dropDuplicates(["OrderID"])
    # Handle missing anomaly type
    orders = orders.withColumn(
        "AnomalyType",
        F.when((F.col("AnomalyFlag") == 1) & (F.col("AnomalyType").isNull()), "Unknown")
        .when((F.col("AnomalyFlag") == 1) & (F.col("AnomalyType") == ""), "Unknown")
    )

```

```

        .otherwise(F.col("AnomalyType"))
    )
    # Add business rule fields
    orders = (
        orders.withColumn("IsAnomaly", F.col("AnomalyFlag") == 1)
        .withColumn("DeviceStatus", F.upper(F.col("DeviceStatus")))
        .withColumn("NormalizedStoreID", F.upper(F.col("StoreID")))
        .withColumn("NormalizedProduct", F.upper(F.col("Product")))
    )
    return orders

```

(iv) Gold Layer

Gold Layer Tables (capstone_retail.gold)

1. Daily / Monthly Sales per Region

```

@table(
    name="cap1_shiv_catalog.capstone_retail_shivankur.gold_sales_region",
    comment="Daily & Monthly Sales per Region"
)
def gold_sales_region():
    return (
        dlt.read("cap1_shiv_catalog.capstone_retail_shivankur.silver_orders")
        .withColumn("OrderDate", F.to_date("OrderTime", "M/d/yyyy H:mm"))
        .withColumn("YearMonth", F.date_format("OrderDate", "yyyy-MM"))
        .groupBy("Region", "OrderDate", "YearMonth")
        .agg(F.sum("SalesAmount").alias("TotalSales"))
    )

```

2. Device Anomaly Trend per Store

```

@table(
    name="cap1_shiv_catalog.capstone_retail_shivankur.gold_anomaly_trends",
    comment="Device Anomaly Trend per Store"
)
def gold_anomaly_trends():
    return (
        dlt.read("cap1_shiv_catalog.capstone_retail_shivankur.silver_orders")
        .withColumn("OrderDate", F.to_date("OrderTime", "M/d/yyyy H:mm"))
        .groupBy("StoreID", "OrderDate", "DeviceType", "AnomalyType")
        .agg(F.count("*").alias("AnomalyCount"))
    )

```

3. Conversion Rate Impact when devices fail

```

@table(
    name="cap1_shiv_catalog.capstone_retail_shivankur.gold_conversion_impact",
    comment="Conversion rate impact of device failures"
)
def gold_conversion_impact():
    df = dlt.read("cap1_shiv_catalog.capstone_retail_shivankur.silver_orders")
    total_orders = df.groupBy("StoreID").agg(F.count("*").alias("TotalOrders"))
    failed_orders = (
        df.filter(df.AnomalyFlag == 1)

```

```

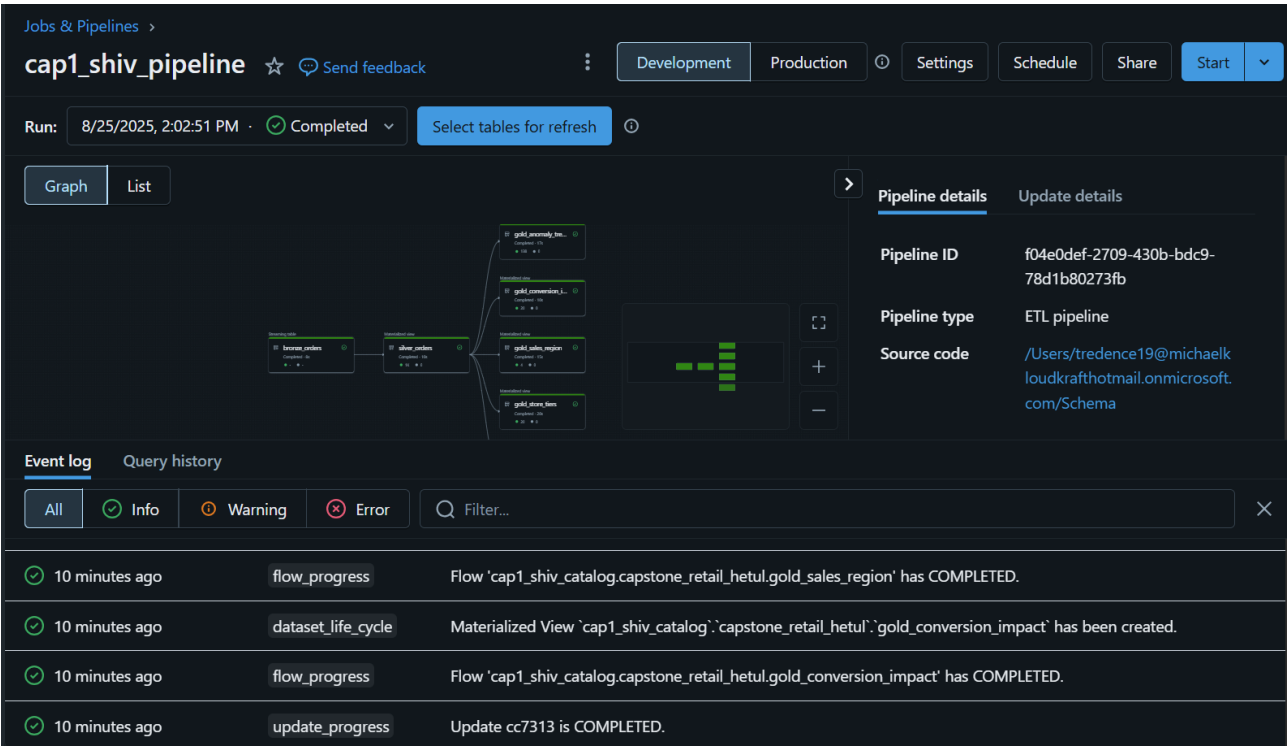
        .groupBy("StoreID")
        .agg(F.count("*").alias("FailedOrders"))
    )
    return (
        total_orders.join(failed_orders, "StoreID", "left")
        .withColumn("FailedOrders", F.coalesce(F.col("FailedOrders"), F.lit(0)))
        .withColumn("FailureRate", F.col("FailedOrders") / F.col("TotalOrders"))
    )
}

# 4. Store Tier Classification
@table(
    name="cap1_shiv_catalog.capstone_retail_shivankur.gold_store_tiers",
    comment="Store tiers based on total sales (High / Medium / Low performers)"
)
def gold_store_tiers():
    df = (
        dlt.read("cap1_shiv_catalog.capstone_retail_shivankur.silver_orders")
        .groupBy("StoreID")
        .agg(F.sum("SalesAmount").alias("TotalSales"))
    )
    return (
        df.withColumn(
            "StoreTier",
            F.when(F.col("TotalSales") > 100000, "High")
            .when(F.col("TotalSales") > 50000, "Medium")
            .otherwise("Low")
        )
    )
}

# 5. Weighted Average Sales vs Anomalies
@table(
    name="cap1_shiv_catalog.capstone_retail_shivankur.gold_weighted_sales_anomalies",
    comment="Weighted average of sales vs anomalies per store"
)
def gold_weighted_sales_anomalies():
    df = dlt.read("cap1_shiv_catalog.capstone_retail_shivankur.silver_orders")
    return (
        df.groupBy("StoreID")
        .agg(
            F.sum("SalesAmount").alias("TotalSales"),
            F.sum(F.when(F.col("AnomalyFlag") == 1, 1).otherwise(0)).alias("TotalAnomalies")
        )
        .withColumn("SalesPerAnomaly",
            F.when(F.col("TotalAnomalies") > 0, F.col("TotalSales") / F.col("TotalAnomalies"))
            .otherwise(F.lit(None))
        )
    )
}

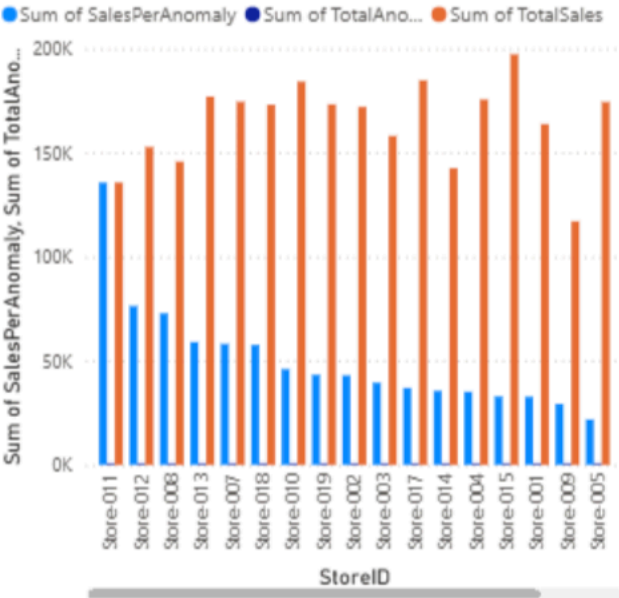
```

Final Pipeline



Power BI

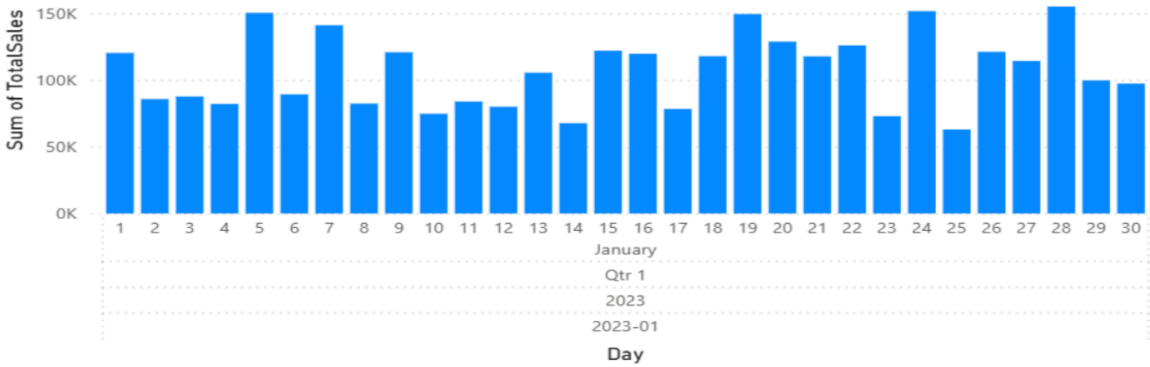
Sum of SalesPerAnomaly, Sum of TotalAnomalies and Sum of TotalSales by StoreID



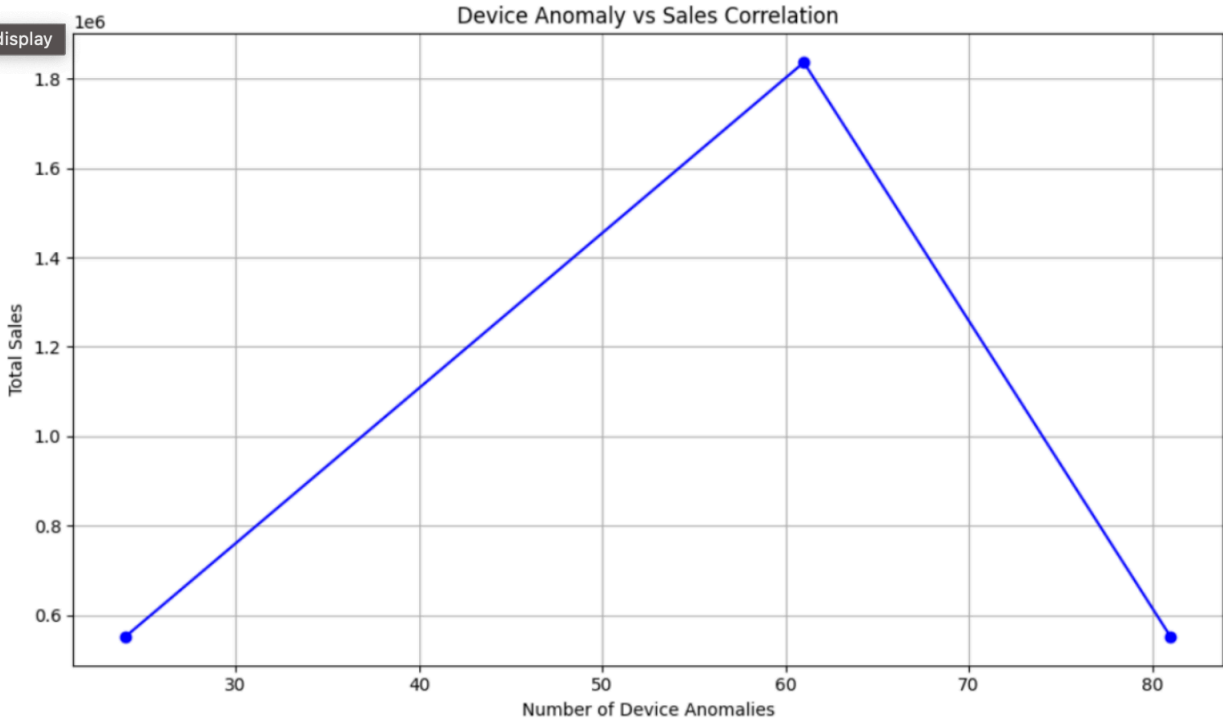
StoreID	StoreTier	Sum of TotalSales
Store-001	High	1,63,738.09
Store-002	High	1,71,989.14
Store-003	High	1,58,060.91
Store-004	High	1,75,558.23
Store-005	High	1,74,334.11
Store-006	High	1,47,107.99
Store-007	High	1,74,436.47
Store-008	High	1,45,685.27
Store-009	High	1,17,012.92
Store-010	High	1,84,143.77
Store-011	High	1,35,648.97
Store-012	High	1,52,687.06
Store-013	High	1,76,925.59
Store-014	High	1,42,545.46
Store-015	High	1,97,270.28
Store-016	High	1,74,728.79
Store-017	High	1,84,736.13
Store-018	High	1,73,029.89
Store-019	High	1,73,193.32
Store-020	Medium	83,245.81
Total		32,06,078.20

Sum of TotalSales by YearMonth, Year, Quarter, Month and Day

File display



File display



2. Event Hub

Code:

```
# =====
# Bronze → Silver → Gold with Event Hub Integration
# =====
from dlt import table, read
import pyspark.sql.functions as F
from pyspark.sql.types import *
# =====
# :one: Bronze Layer: Schema & CSV
# =====
iot_schema = StructType([
    StructField("OrderID", StringType(), True),
    StructField("CustomerID", StringType(), True),
    StructField("Region", StringType(), True),
    StructField("StoreID", StringType(), True),
    StructField("Product", StringType(), True),
    StructField("Quantity", IntegerType(), True),
    StructField("UnitPrice", DoubleType(), True),
    StructField("SalesAmount", DoubleType(), True),
    StructField("OrderTime", StringType(), True),
    StructField("DeviceType", StringType(), True),
    StructField("Temperature", DoubleType(), True),
    StructField("DeviceStatus", StringType(), True),
    StructField("AnomalyFlag", IntegerType(), True),
    StructField("AnomalyType", StringType(), True)
])
raw_path = "abfss://raw-1@shivghaie.dfs.core.windows.net/incoming/"
schema_path = "abfss://raw-1@shivghaie.dfs.core.windows.net/dlt_schemas/bronze_orders"
@table(
    name="retail_catalog_shivankur.bronze.bronze_orders_csv",
    comment="Bronze table - raw IoT CSV"
)
def bronze_orders_csv():
    return (
        spark.readStream
            .format("cloudFiles")
            .option("cloudFiles.format", "csv")
            .option("header", True)
            .option("cloudFiles.schemaLocation", schema_path)
            .schema(iot_schema)
            .load(raw_path)
            .withColumn("OrderTime", F.to_timestamp("OrderTime"))
    )
# =====
# :two: Bronze Layer: Event Hub IoT
# =====
eventhub_connection_string = ""
```



```

consumer_group = "$Default"
ehConf = {
    'eventhubs.connectionString': eventhub_connection_string,
    'eventhubs.consumerGroup': consumer_group,
    'eventhubs.startingPosition': '@latest'
}
@table(
    name="retail_catalog_shivankur.bronze.bronze_orders_eventhub",
    comment="Bronze table - live IoT data from Event Hub"
)
def bronze_orders_eventhub():
    df = spark.readStream \
        .format("eventhubs") \
        .options(**ehConf) \
        .load()
    # Event Hub body is bytes → convert to string → parse JSON
    df1 = df.selectExpr("cast(body as string) as json_string")
    df_parsed = df1.select(F.from_json(F.col("json_string"), iot_schema).alias("data")).select("data.*")
    return df_parsed.withColumn("OrderTime", F.to_timestamp("OrderTime"))
# =====
# :three: Unified Bronze Table
# =====
@table(
    name="cap1_shiv_catalog.capstone_retail_shivankur.bronze_orders",
    comment="Unified Bronze table (CSV + Event Hub)"
)
def bronze_orders():
    df_csv = read("cap1_shiv_catalog.capstone_retail_shivankur.bronze_orders_csv")
    df_eh = read("cap1_shiv_catalog.capstone_retail_shivankur.bronze_orders_eventhub")
    return df_csv.unionByName(df_eh)
# =====
# :four: Silver Layer
# =====
@table(
    name="cap1_shiv_catalog.capstone_retail_shivankur.silver_orders",
    comment="Silver layer: cleaned and deduplicated IoT orders with anomaly handling"
)
def silver_orders():
    orders = read("cap1_shiv_catalog.capstone_retail_shivankur.bronze_orders")
    orders = (
        orders.withColumn("Quantity", F.col("Quantity").cast(IntegerType()))
        .withColumn("UnitPrice", F.col("UnitPrice").cast(DoubleType()))
        .withColumn("SalesAmount", F.col("SalesAmount").cast(DoubleType()))
        .withColumn("Temperature", F.col("Temperature").cast(DoubleType()))
        .withColumn("OrderTime", F.to_timestamp("OrderTime", "M/d/yyyy H:mm"))
    )
    orders = orders.dropDuplicates(["OrderID"])
    orders = orders.withColumn(
        "AnomalyType",
        F.when((F.col("AnomalyFlag") == 1) & (F.col("AnomalyType").isNull()), "Unknown")
        .when((F.col("AnomalyFlag") == 1) & (F.col("AnomalyType") == ""), "Unknown")
    )

```

```

        .otherwise(F.col("AnomalyType"))
    )
    orders = (
        orders.withColumn("IsAnomaly", F.col("AnomalyFlag") == 1)
        .withColumn("DeviceStatus", F.upper(F.col("DeviceStatus")))
        .withColumn("NormalizedStoreID", F.upper(F.col("StoreID")))
        .withColumn("NormalizedProduct", F.upper(F.col("Product")))
    )
    return orders
# =====
# :five: Gold Layer
# =====
@table(
    name="cap1_shiv_catalog.capstone_retail_shivankur.gold_sales_region",
    comment="Daily & Monthly Sales per Region"
)
def gold_sales_region():
    return (
        read("cap1_shiv_catalog.capstone_retail_shivankur.silver_orders")
        .withColumn("OrderDate", F.to_date("OrderTime", "M/d/yyyy H:mm"))
        .withColumn("YearMonth", F.date_format("OrderDate", "yyyy-MM"))
        .groupBy("Region", "OrderDate", "YearMonth")
        .agg(F.sum("SalesAmount").alias("TotalSales"))
    )
@table(
    name="cap1_shiv_catalog.capstone_retail_shivankur.gold_anomaly_trends",
    comment="Device Anomaly Trend per Store"
)
def gold_anomaly_trends():
    return (
        read("cap1_shiv_catalog.capstone_retail_shivankur.silver_orders")
        .withColumn("OrderDate", F.to_date("OrderTime", "M/d/yyyy H:mm"))
        .groupBy("StoreID", "OrderDate", "DeviceType", "AnomalyType")
        .agg(F.count("*").alias("AnomalyCount"))
    )
@table(
    name="cap1_shiv_catalog.capstone_retail_shivankur.gold_conversion_impact",
    comment="Conversion rate impact of device failures"
)
def gold_conversion_impact():
    df = read("cap1_shiv_catalog.capstone_retail_shivankur.silver_orders")
    total_orders = df.groupBy("StoreID").agg(F.count("*").alias("TotalOrders"))
    failed_orders = (
        df.filter(df.AnomalyFlag == 1)
        .groupBy("StoreID")
        .agg(F.count("*").alias("FailedOrders"))
    )
    return (
        total_orders.join(failed_orders, "StoreID", "left")
        .withColumn("FailedOrders", F.coalesce(F.col("FailedOrders"), F.lit(0)))
        .withColumn("FailureRate", F.col("FailedOrders") / F.col("TotalOrders"))
    )

```

```

@table(
  name="cap1_shiv_catalog.capstone_retail_shivankur.gold_store_tiers",
  comment="Store tiers based on total sales (High / Medium / Low performers)"
)
def gold_store_tiers():
  df = (
    read("cap1_shiv_catalog.capstone_retail_shivankur.silver_orders")
    .groupBy("StoreID")
    .agg(F.sum("SalesAmount").alias("TotalSales"))
  )
  return (
    df.withColumn(
      "StoreTier",
      F.when(F.col("TotalSales") > 100000, "High")
      .when(F.col("TotalSales") > 50000, "Medium")
      .otherwise("Low")
    )
  )
@table(
  name="cap1_shiv_catalog.capstone_retail_shivankur.gold_weighted_sales_anomalies",
  comment="Weighted average of sales vs anomalies per store"
)
def gold_weighted_sales_anomalies():
  df = read("cap1_shiv_catalog.capstone_retail_shivankur.silver_orders")
  return (
    df.groupBy("StoreID")
    .agg(
      F.sum("SalesAmount").alias("TotalSales"),
      F.sum(F.when(F.col("AnomalyFlag") == 1, 1).otherwise(0)).alias("TotalAnomalies")
    )
    .withColumn("SalesPerAnomaly",
      F.when(F.col("TotalAnomalies") > 0, F.col("TotalSales") / F.col("TotalAnomalies"))
      .otherwise(F.lit(None))
    )
  )

```

Final Output:

Microsoft Azure databricks

Search data, notebooks, recent, and more...

tred_d5

New

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie

Alerts

Query History

SQL Warehouses

Data Engineering

Job Runs

Data Ingestion

AI/ML

Playground

Experiments

Features

Models

Serving

Jobs & Pipelines

shiv_eventhub_pipeline

Development

Production

Settings

Schedule

Share

Start

Run: 25/8/2025, 9:16:13 PM

Failed

Select tables for refresh

Creating update

Waiting for resources

Initializing

4 Setting up tables

5 Rendering graph

Pipeline details

Update details

Pipeline ID

8d5d211a-5180-445f-bab9-fb68b7e8a309

Pipeline type

ETL pipeline

Source code

/Users/tredence19@michaelkloudkrafthotmail.onmicrosoft.com/Untitled Notebook 2025-08-25 20:43:58

Run as

Tredence LabsKraft19

Tags

None

Event log

Query history

All

Info

Warning

Error

Filter...

17 seconds ago

flow_progress

Failed to resolve flow due to upstream failure: 'cap1_shiv_catalog.capstone_retail_hetul.gold_store_tiers'.

17 seconds ago

flow_progress

Failed to resolve flow: 'retail_catalog_hetul.bronze.bronze_orders_eventhub'.

17 seconds ago

flow_progress

Failed to resolve flow: 'cap1_shiv_catalog.capstone_retail_hetul.bronze_orders'.

17 seconds ago

update_progress

Update 5e47fe has failed. Failed to analyze flow 'retail_catalog_hetul.bronze.bronze_orders_eventhub' and 1 other flow(s)...

Terminal

nano iot_stream.py

```
GNU nano 7.2 iot_stream.py
import random
import time
import uuid
from datetime import datetime
from azure.eventhub import EventHubProducerClient, EventData

# CONFIGURATION
EVENT_HUB_CONNECTION_STR = "Endpoint=sb://shivghaie.servicebus.windows.net;/SharedAccessKeyName=shive;SharedAccessKeys=VW+9ic1Thp2scj66h8oDqa+WIf6jy0fE+AEhCVR5n4n;EntityPath=shiveventhub"
EVENT_HUB_NAME = "shiveventhub"

# Function to generate one IoT row
def generate_row():
    quantity = random.randint(1, 10)
    unit_price = round(random.uniform(50, 500), 2)
    sales_amount = round(quantity * unit_price, 2)
    device_status = random.choice(["Active", "Inactive", "Error"])
    anomaly_flag = 1 if device_status != "Active" else 0
    anomaly_type = random.choice(["Overheat", "LowBattery", "Disconnected"]) if anomaly_flag else "None"
    return {
        "OrderID": str(uuid.uuid4()),
        "CustomerID": f"CUST{random.randint(1000, 9999)}",
        "Region": random.choice(["North", "South", "East", "West"]),
        "StoreID": random.choice(["Store001", "Store002", "Store003"]),
        "Product": random.choice(["Laptop", "Phone", "Monitor", "Keyboard", "Mouse"]),
        "Quantity": quantity,
        "UnitPrice": unit_price,
        "SalesAmount": sales_amount,
        "OrderTime": datetime.now().strftime("%Y-%m-%d %H:%M:%S"),
        "DeviceType": random.choice(["SensorA", "SensorB", "SensorC"]),
        "Temperature": round(random.uniform(20, 100), 2),
        "DeviceStatus": device_status,
        "AnomalyFlag": anomaly_flag,
        "AnomalyType": anomaly_type
    }

# Event Hub Producer
producer = EventHubProducerClient.from_connection_string(
    conn_str=EVENT_HUB_CONNECTION_STR,
    eventhub_name=EVENT_HUB_NAME
)

# Send data every second
try:
    for _ in range(1000): # Send exactly 1000 rows
        row = generate_row()
        event_data = EventData(str(row)) # convert dict to string
        with producer:
            producer.send_batch([event_data])
        print(f"Sent row: {row}")
        time.sleep(1) # 1-second interval
except KeyboardInterrupt:
    print("Streaming stopped by user.")
finally:
    producer.close()
```